Nishant Gadde

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Education

The University of Texas at Austin

B.S. in Electrical and Computer Engineering (Engineering Honors) and Mathematics *GPA: 3.87/4.0*

- **Coursework:** Probability & Random Processes, Optimization, Reinforcement Learning, Numerical Methods, Complex Analysis, Machine Learning.

Honors & Competitions

USACO Platinum Division (∼top 1%), USA Computing Olympiad **2024–2025**

- Ranked **top 100 worldwide**; built strong foundations in search, planning, and adversarial problem-solving.

Experience

Machine Learning Research Intern (LLM Infrastructure) Jun 2026 – Aug 2026

Apple *Seattle, WA*

- Optimized batching and caching strategies for large-scale LLM inference workloads, improving GPU utilization and serving efficiency.
- Improved throughput, latency, and reliability across multi-tenant agent services while maintaining 99.99% production uptime.
- Built tooling APIs, observability pipelines, and production infrastructure supporting 50+ internal LLM-powered services.

Research Assistant, Vision–Language Agents Jan 2026 – Present

Oden Institute for Computational Engineering & Sciences, UT Austin *Austin, TX*

- Built vision-language agent architectures for multi-step reasoning and tool use across interactive environments.
- Designed adaptive routing between fast and large models to optimize latency, accuracy, and inference cost.
- Developed evaluation infrastructure with memory, context compression, and automated benchmarking for agent systems.

Machine Learning Research Intern Jun 2025 – Aug 2025

Rayfield Systems *Houston, TX*

- Trained recurrent state-space models for high-frequency processes, powering risk-aware time-series foundation models that outperformed classic baselines.
- Optimized model serving through quantization and streaming inference, reducing memory usage 4×.
- Delivered time-series foundation models & control models with automated eval harnesses, monitoring drift and anomaly patterns for 10+ clients.

Research Assistant, Sustainable Power Systems Jun 2024 – Aug 2025

Stanford Sustainability Systems Laboratory (S3L) *Remote*

- Co-authored IEEE CAI work on ML approximations of AC optimal power flow with <1% gaps and 1000× speedups.
- Integrated generative models with RL agents for scalable decision-making under system constraints.
- Implemented reproducible evaluation pipelines with baselines, ablations, and energy measurements for large-scale agents.

Selected Projects

Multi-Agent Research Assistant for Safe Tool-Use | *Python, REST, memory* **2025–2026**

- Engineered planner, retrieval, execution, and reviewer agents coordinating via persistent memory and task decomposition.
- Implemented context compression, retrieval, citation tracking, and sandboxed tool execution to improve reliability.
- Built automated evaluation pipelines measuring tool-use reliability, citation quality, and planning accuracy.

Tool-Use Fine-Tuning of Hugging Face Agents | *HF Transformers, RLHF* **2025**

- Generated synthetic trajectories for tool-calling agents using Hugging Face Transformers.
- Used preference optimization & reward models to favor interpretable, aligned sequences over brittle single-shot behaviors.
- Benchmarked finetuned agents on code & data tasks, logging win rates, robustness metrics, and failure modes.

Hamiltonian Flow Matching for Robust Grid Control | *PyTorch, RL, generative modeling* **2026**

- Proposed Hamiltonian Flow Matching (HFM), enforcing Kirchhoff-consistent flows for safety-critical power grid scenarios.
- Designed curvature-based schedulers that cut GPU energy per scenario while preserving risk-sensitive forecast accuracy.
- Released GridControl-RL benchmark coupling generation with SAC dispatch agents to study robustness and constraint violations.

Selected Publications

Siva, R.*, Bhatt, N. P.*, Yang, Y., Lee, S., **Gadde, N.**, Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. "What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning," *CoRL*, 2026.

Prakash, S., Cheng, A., Tschand, A., Mazumder, M., Gohil, V., ... **Gadde, N.**, ... Yazdanbakhsh, A., & Janapa Reddi, V. "QuArch: A Benchmark for Evaluating LLM Reasoning in Computer Architecture," *ICML*, 2026.

Miltner, M., Vašata, D., Friedjungová, M., **Gadde, N.**, Štogl, O., Starý, O., & Rajagopal, R. "Towards Real-Time Machine Learning Approximations of AC Optimal Power Flow," *IEEE CAI*, 2025.

Technical Skills

LLM Systems: Inference optimization, agent architectures, tool use, function calling, memory, context compression, evaluation, benchmarking, prompt engineering.

Machine Learning: PyTorch, TensorFlow, Transformers, RL (SAC, PPO), sequence modeling, generative modeling, time-series models.

Systems: Python, C++, Rust, Linux, CUDA, distributed inference, REST APIs, SQL, Kafka, Git.

Practices: Experimentation, evaluation, A/B testing, profiling, benchmarking, reproducible ML.

Math & Quant: Probability & Statistics, Optimization, Stochastic Processes, Linear Algebra, Numerical Analysis.